

Pangenome based genome inference

EMBO Practical Course 2026, Naples



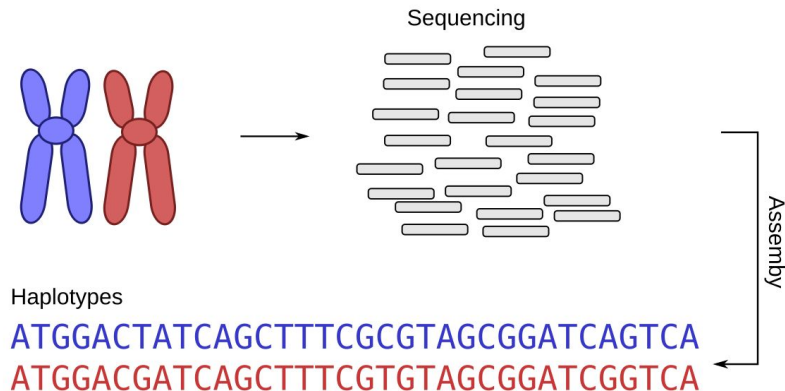
Jana Ebler

PostDoc @ Marschall Lab

Heinrich Heine Universität Düsseldorf, Germany

Haplotype-resolved genome assembly

- New methods allow near T2T assembly of human haplotypes
- based on long read sequencing (PacBio HiFi, ONT)
- still expensive and not yet applicable to large cohorts
- So far, number of assembled haplotypes is relatively low (~600)



Pangenome graphs

Haplotype assemblies enable the construction of **pangenome graphs**

- Multiple references instead of one
- Graph represents variation in a species
- Replace linear reference to improve downstream analyses

Previous human genome reference

VS

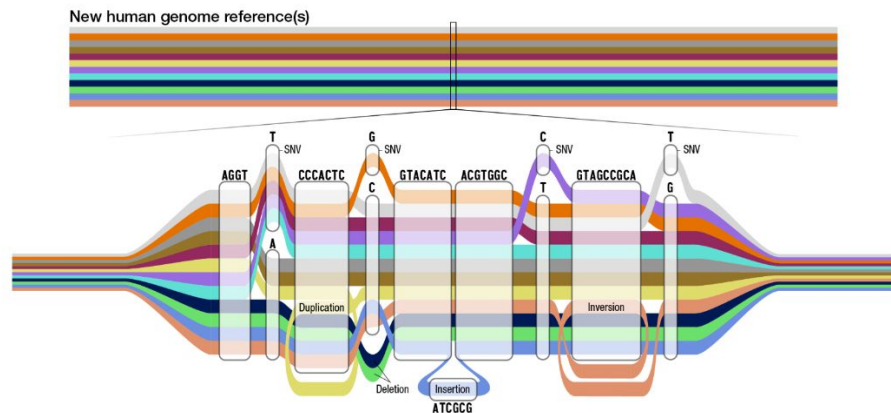
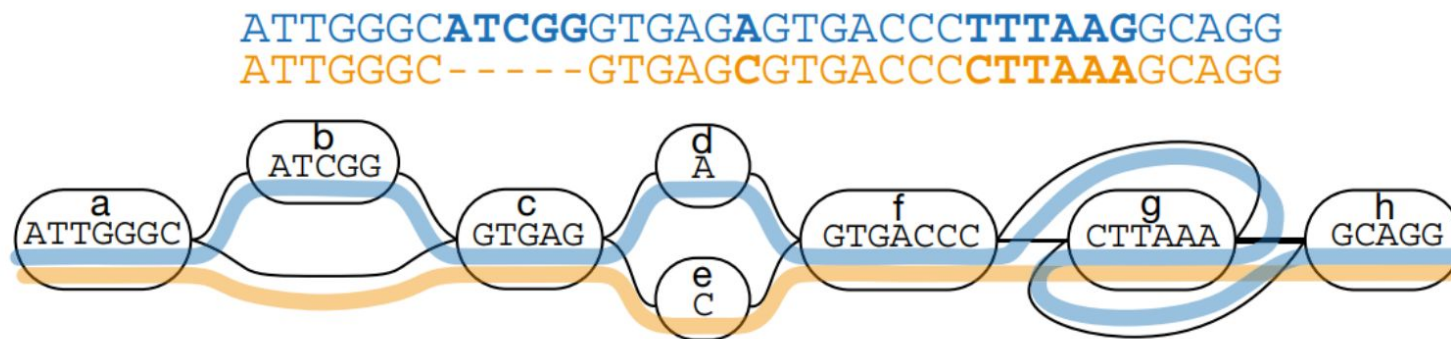


Figure Credit: Darryl Leja, NHGRI

Pangenome graphs



W.-W. Liao, M. Asri, **J. Ebler**, et al. *A draft human pangenome reference*. Nature (2023)

Using Pangenomes to interpret short read data sets



J. Ebler, P. Ebert, W. E. Clarke, T. Rausch, P. A. Audano, T. Houwaart, Y. Mao, J. O. Korbel, E. E. Eichler, M. C. Zody, et al.

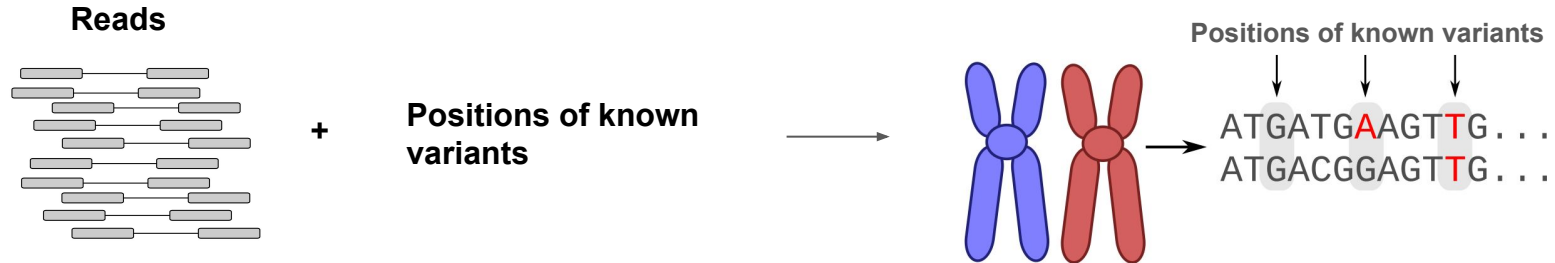
Pangenome-based genome inference allows efficient and accurate genotyping across a wide spectrum of variant classes.

Nature Genetics, 54(4):518–525, 2022



What is genotyping?

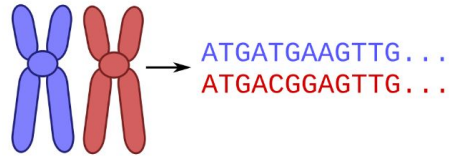
Genotyping: Determine whether none, one or both chromosomes carry a variant



Note: here, we consider the diploid case, as we mostly work with human data

Types of genetic variation

homologous copies



types of genetic variation

a ACGAAGTTGCATGAAAGTATC
ACGGAGTTGCACGAAAGTATC

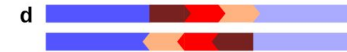
SNP



Deletion



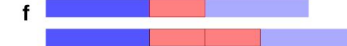
Insertion



Inversion



Interspersed
Duplication

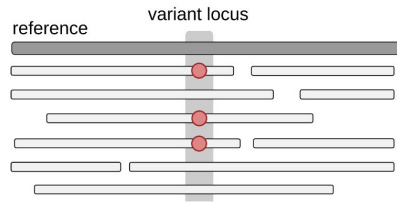


Tandem
Duplication

variants ≥ 50 bp are called **structural variants (SVs)**

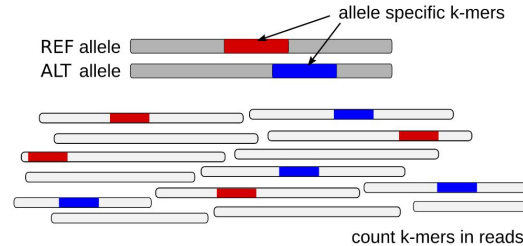
Short-read based genotyping

- Due to low cost, **short-read** sequencing widely used when studying large cohorts
- **Existing genotyping approaches:**



Mapping-based

slow, alignment issues for SVs + repeat regions



K-mer based

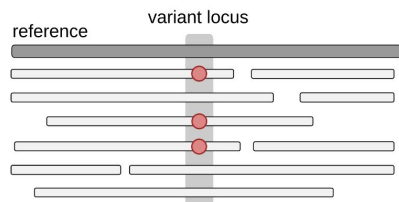
fast, but SVs often located in repeat regions

k-mer: sequence of fixed length k

- **Consequence:**
 - **Short-read approaches miss more than half of all SVs!**
 - Missed variants inaccessible to downstream analyses (e.g. GWAS)
 - Previously, SVs have been mostly excluded from short-read studies

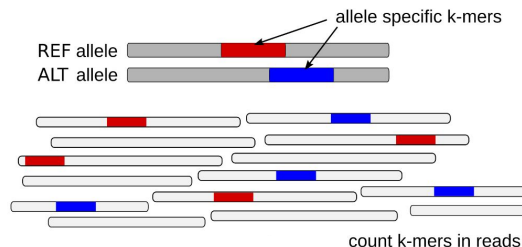
Short-read based genotyping

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- **Existing genotyping approaches:**



Mapping-based

slow, alignment issues for SVs + repeat regions



K-mer based

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k-mer: sequence of fixed length k

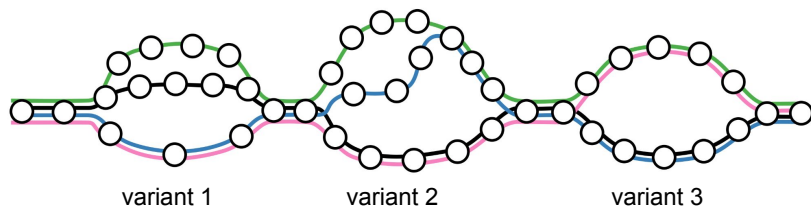
- **Consequence:**
 - **Short-read approaches miss more than half of all SVs!**
 - Missed variants inaccessible to downstream analyses (e.g. GWAS)
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Can pangenomes help to interpret genomes based on short reads?

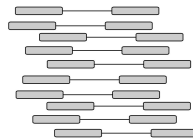
PanGenie: Pangenome-based genotyping

Input:

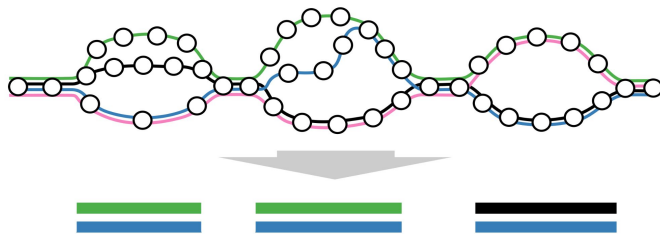
1. **Pangenome graph** constructed from known haplotypes

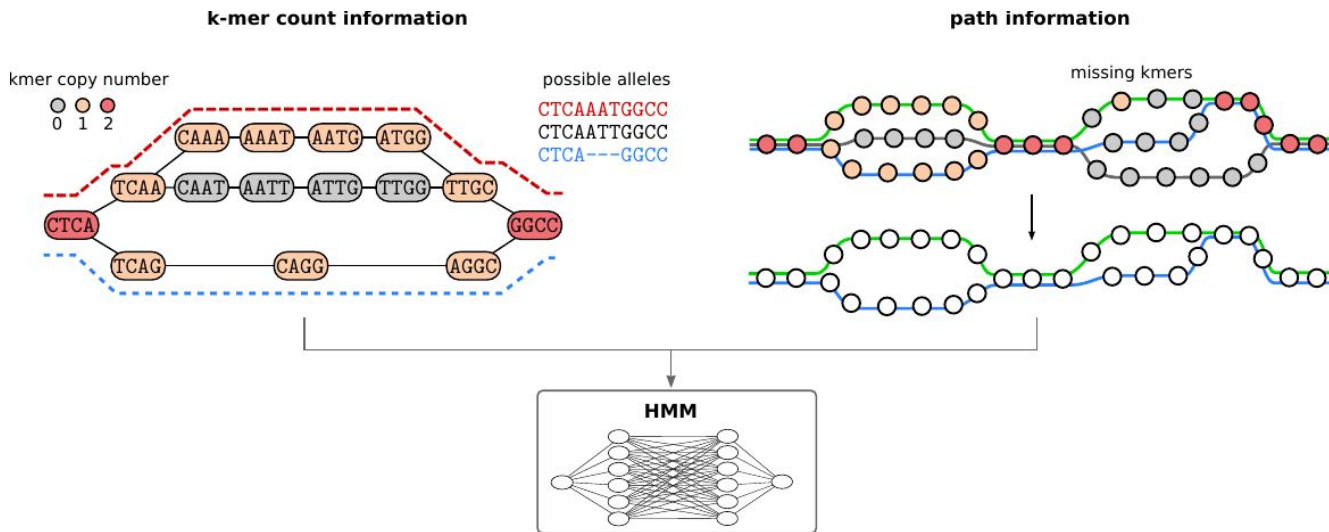


2. **Short reads** of the sample to be genotyped (not in graph!)



Goal: Determine genotype of unknown genome for all variants contained in graph



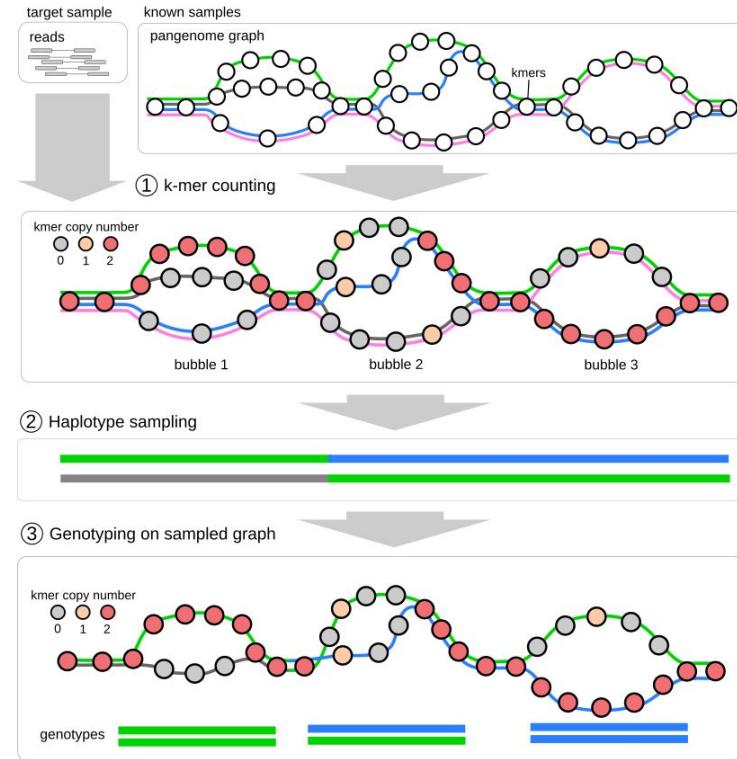


Approach: find two haplotypes that: **(a)** best explain the observed k-mer counts and **(b)** are mosaics of the known haplotypes present in the graph



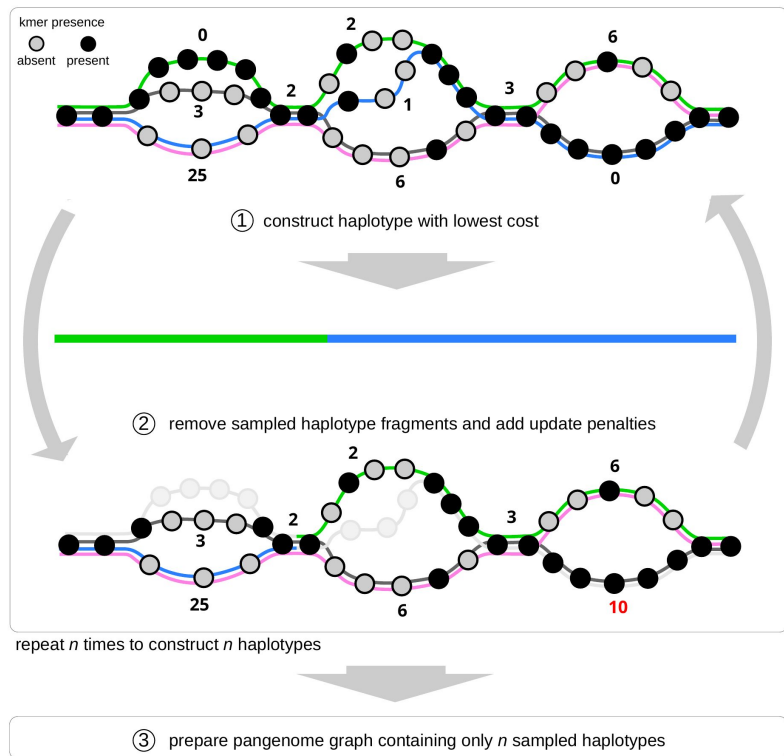
PanGenie: Overview

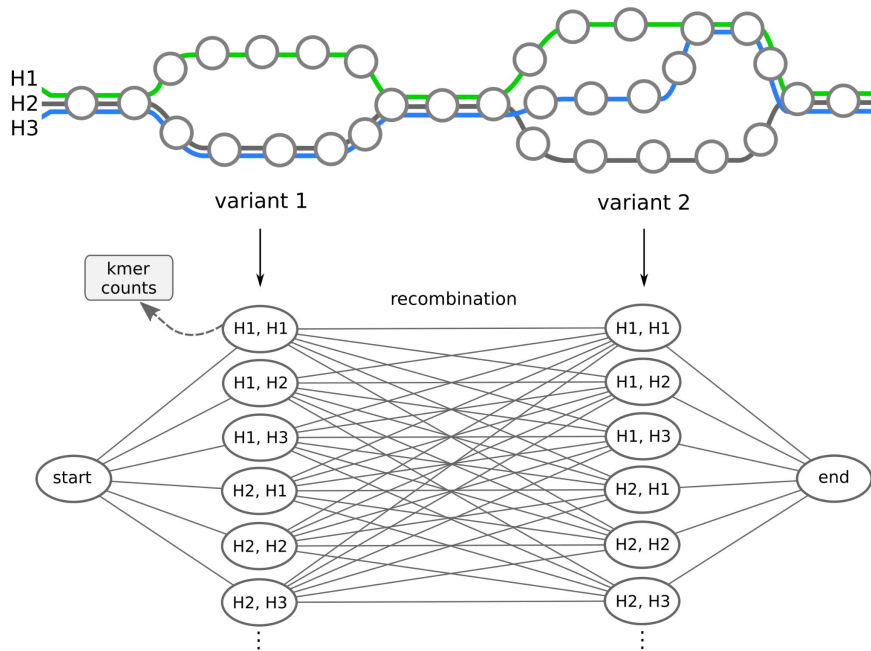
- Determine unique k-mers + count in reads
- Sample small set of haplotypes from full set
- Genotype sample based on:
 - k-mer information and
 - sampled haplotype paths



PanGenie: Haplotype sampling

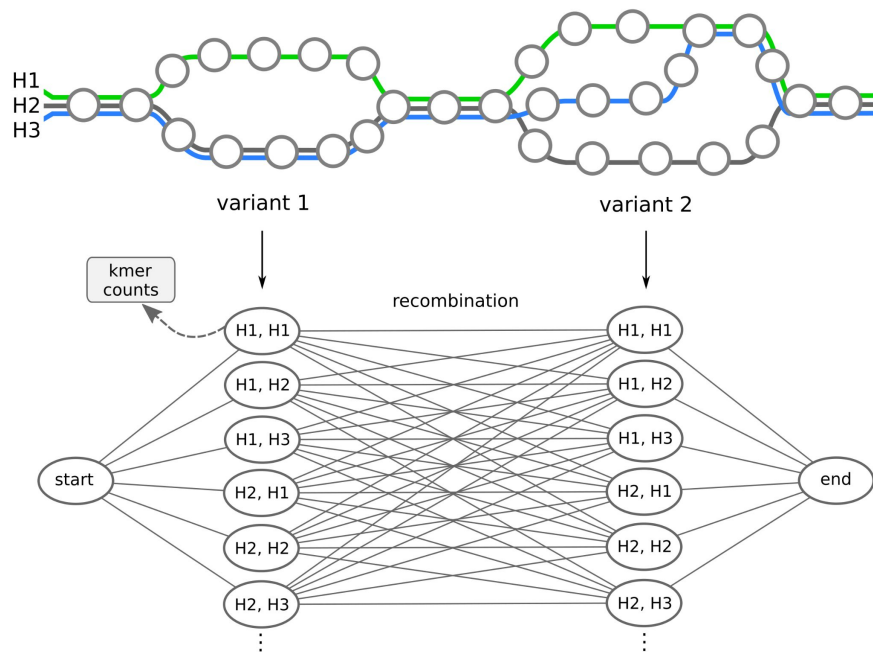
- Speed up genotyping by reducing number of paths in graph
- Genotyping then only based on these paths
- Iterative algorithm that constructs paths supported by sample's read k-mers





Hidden states

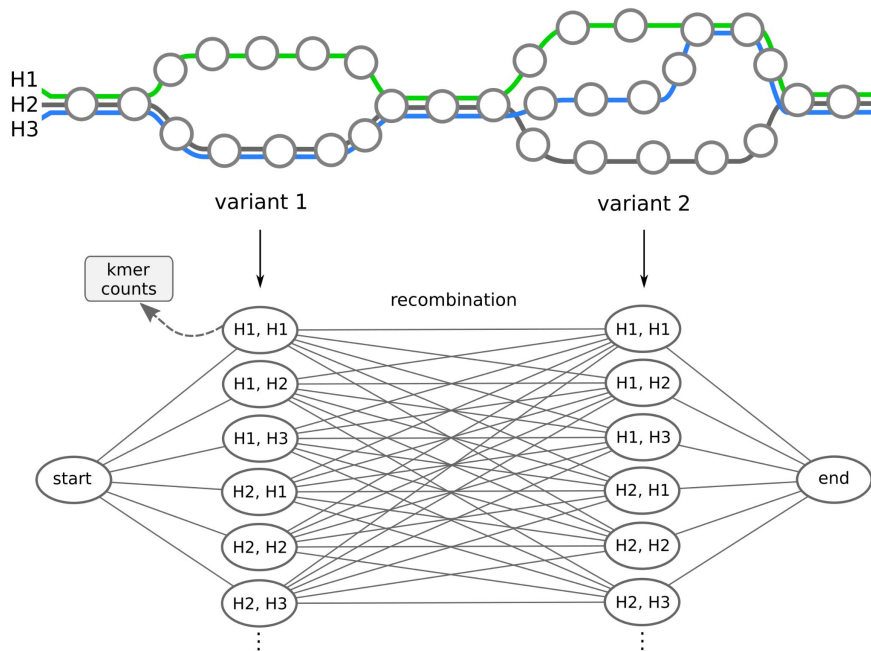
- Pairs of haplotype paths at each position
- Represent possible genotypes of the sample



Emissions:

- States emit counts for k-mers of variant bubble
- Emission probabilities based on expected k-mer coverage

PanGenie: Genotyping



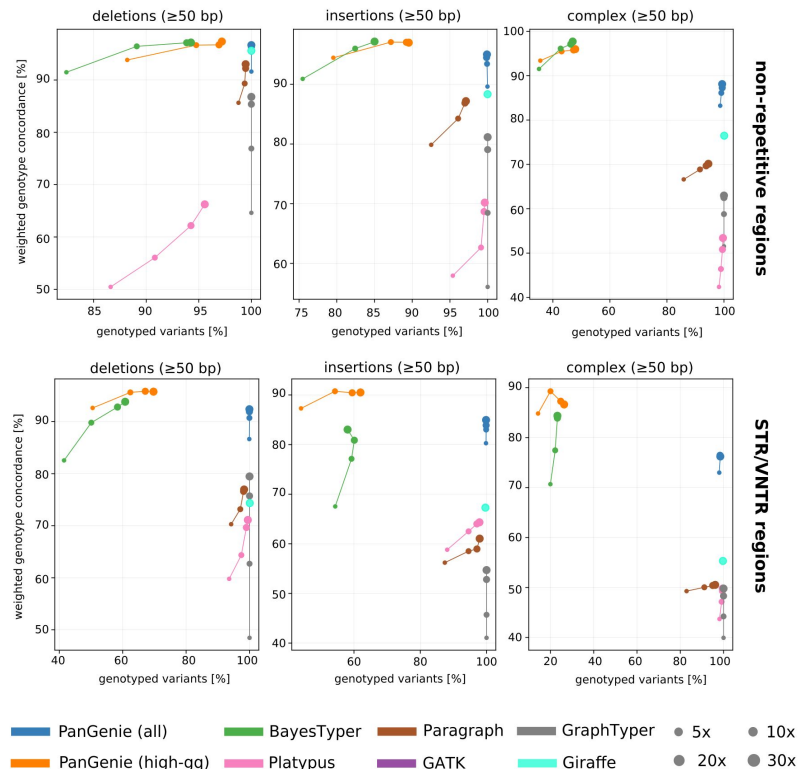
Transitions:

- Represent recombination events
- Probabilities derived from Li-Stephans model
- Probability to switch haplotypes gets higher as distance between variants gets larger

Forward-Backward algorithm is used to compute likeliest genotype at each position

Comparison to linear reference based SV genotypers

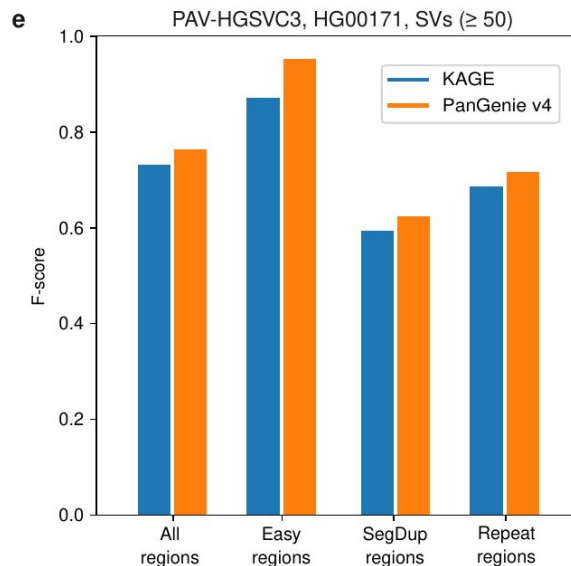
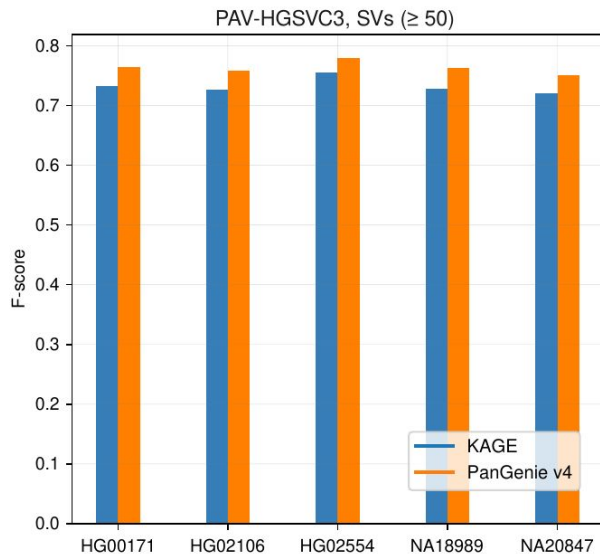
genotyping accuracy for NA12878 (SVs)



- **PanGenie outperforms SV genotypers relying on linear reference genome**
- **Faster than alignment-based methods**

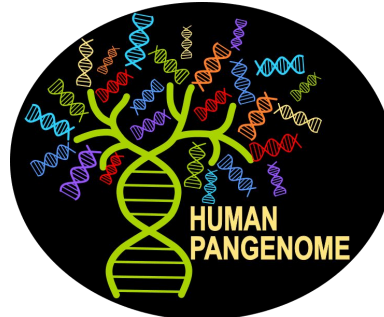
(results for PanGenie v1)

Comparison to pangenome-based approaches



PanGenie performs better than KAGE (v1), especially in “easy” regions of the genome

Application: Human Pangenome Reference Consortium (HPRC2)

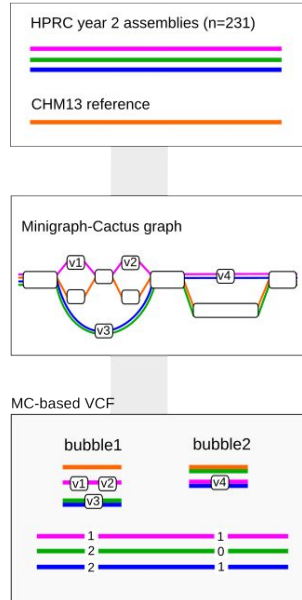


HPRC Release 2 haplotype-resolved assemblies

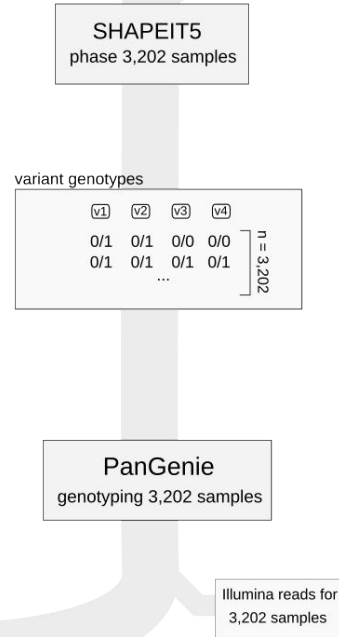
- Assembled with PacBio HiFi, ONT UL and Hi-C sequencing data
- based on Hifiasm assembly with trio or Hi-C phasing
- Assemblies for **231 human samples (462 haplotypes)**

HPRC2 analysis

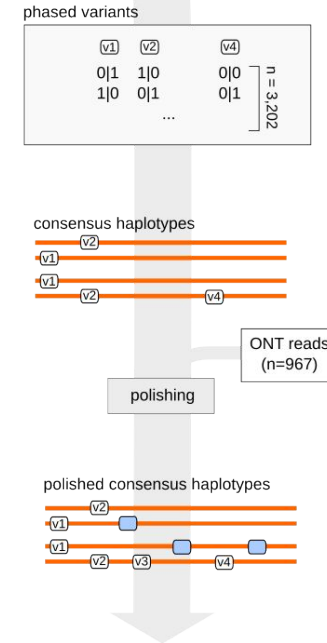
Step 1: graph construction



Step 2: genotyping

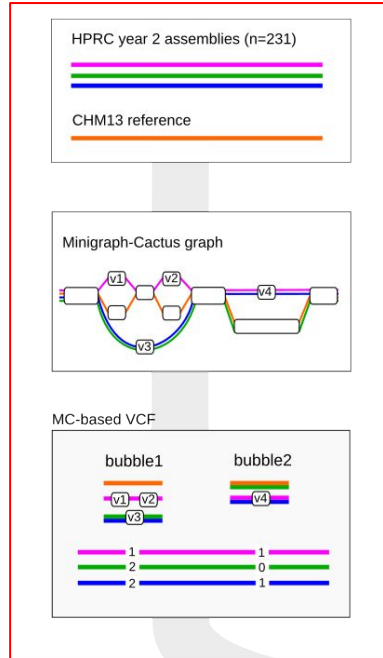


Step 3: haplotype reconstruction

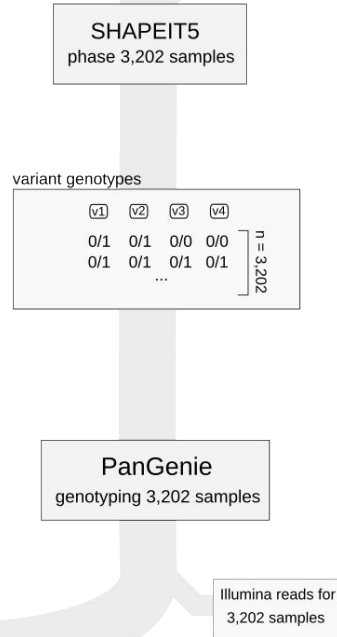


HPRC2 analysis

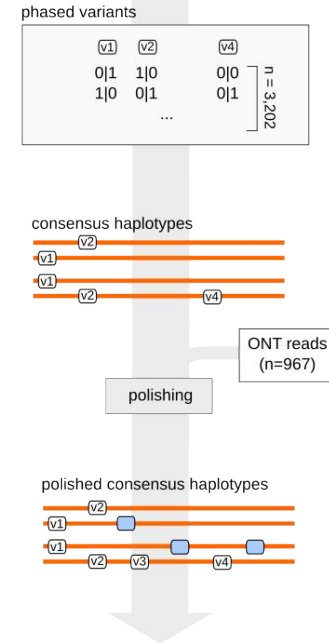
Step 1: graph construction



Step 2: genotyping

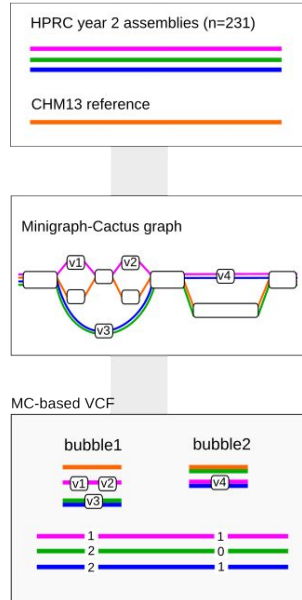


Step 3: haplotype reconstruction

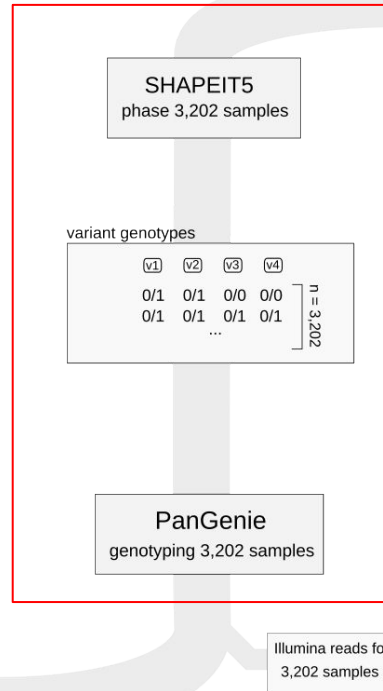


HPRC2 analysis

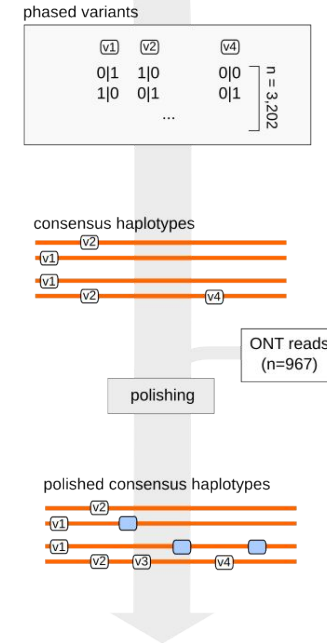
Step 1: graph construction



Step 2: genotyping



Step 3: haplotype reconstruction

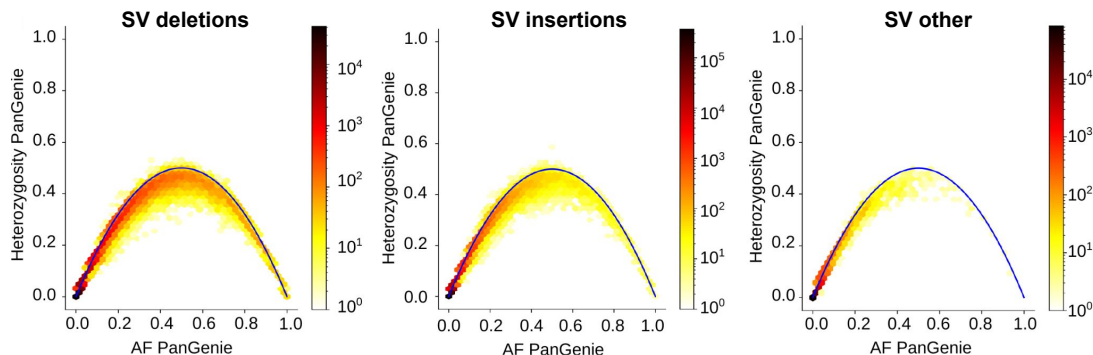


- Variant alleles called from graph

SNPs	Indels (1-49 bp)	SVs (≥ 50 bp)
36,483,743	10,512,438	901,029

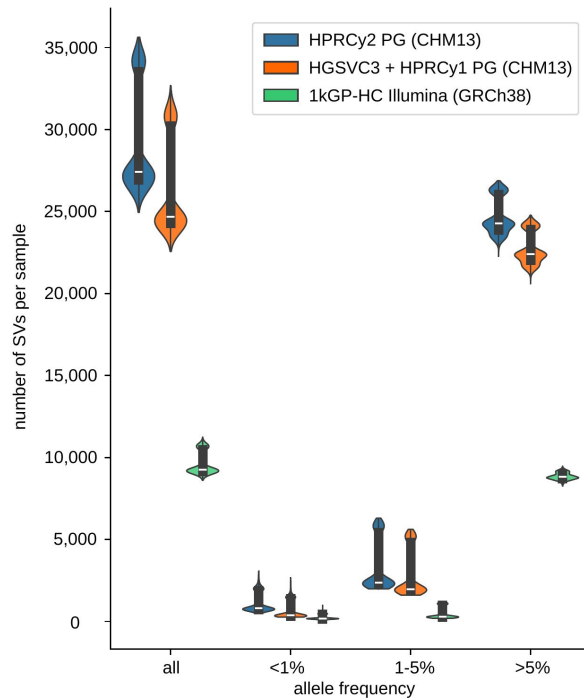
- **Genotyping of these variants across 3,202 human samples from the 1000 Genomes Project with PanGenie**

Genotyping 3,202 samples of the 1kGP cohort



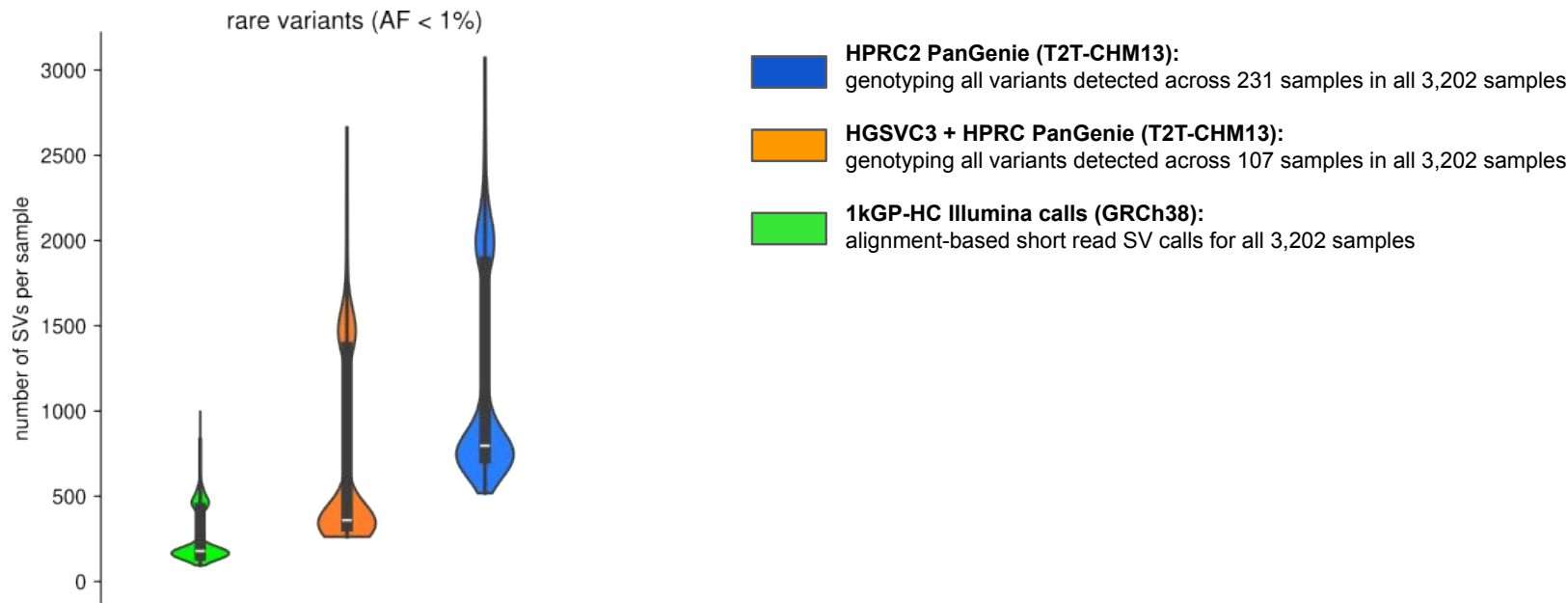
Genotyping based on HPRC2 pangenome with PanGenie, SVs only

- Genotyping in line with Hardy-Weinberg Equilibrium
- PanGenie can access thousands of additional SVs compared to traditional short-read based SV callers



Genotyping with PanGenie based on HPRC2 & HGSVC3 pangenomes, as well as traditional alignment-based SV calling workflow (green)

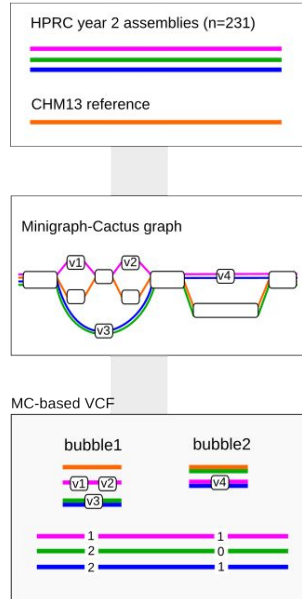
HPRC2 genotyping: rare variants



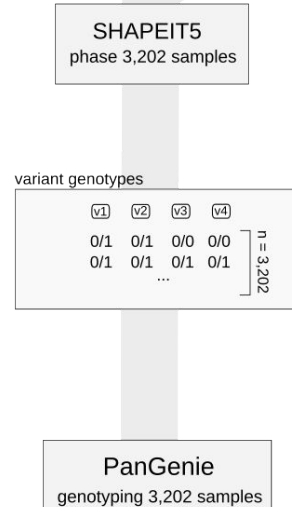
Can access much rare SVs now!

HPRC2 analysis

Step 1: graph construction

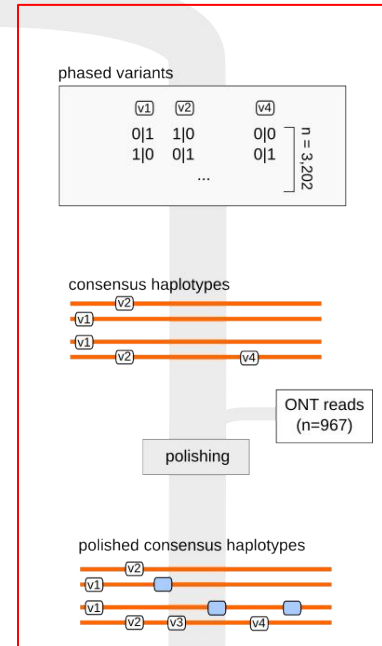


Step 2: genotyping



Illumina reads for
3,202 samples

Step 3: haplotype reconstruction



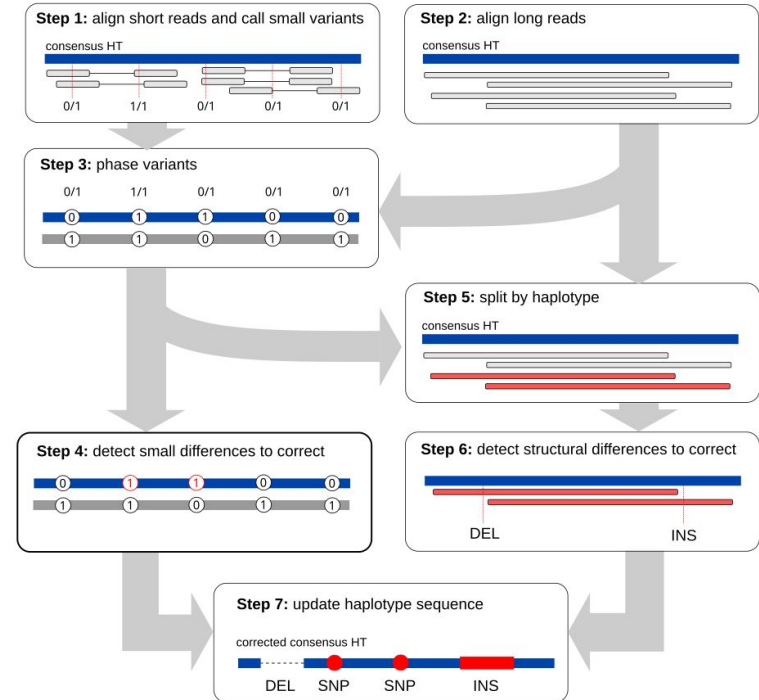
Consensus haplotypes still contain errors

- Genotyping is not perfect
- Genotyping only captures variants present in the pangenome, i.e. in the 462 assembled haplotypes represented in the graph
 - Variants unique to the sample (mainly rare variants) are missed
- **Goal:** correct errors and include missed variants based on:
 - short read data
 - low coverage ONT data

Polishing pipeline

- Polish each consensus haplotype separately
- **Challenge:**
 - Reads are not split by haplotype
 - Need to distinguish true differences from calls originating from opposite haplotype
- **Idea:**
 - Split long reads based on phased small variant calls
 - Detect calls with different phase to find true small variant differences (Step 4)

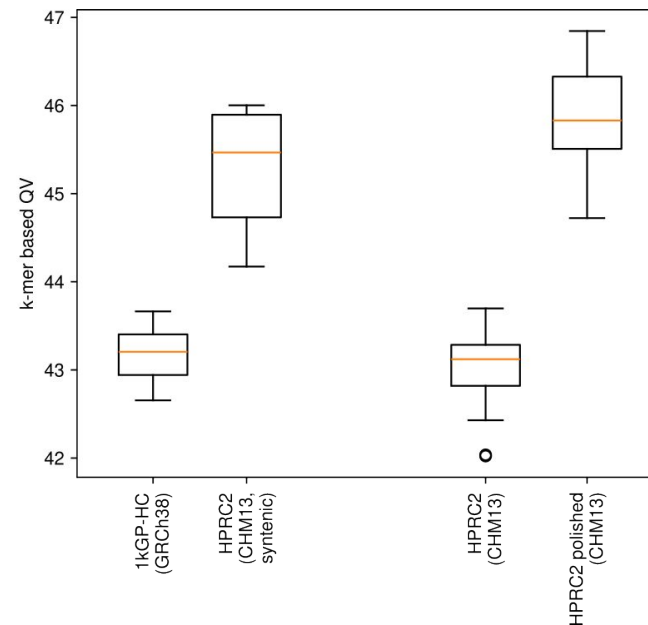
a polishing pipeline



Evaluation of (polished) consensus haplotypes

How accurate are (un)polished haplotypes?

- Applied polishing to all 967 ONT samples
- Evaluated 10 samples (20 haplotypes)
- Computed QV estimates using Merqurey
 - Compares k-mer sets of Illumina reads and predicted haplotype

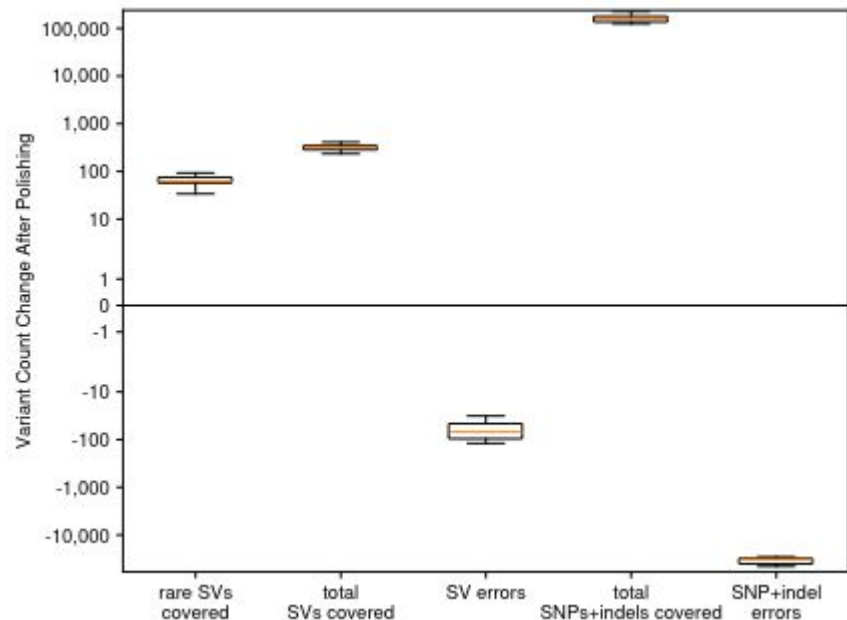


More accurate consensus haplotypes and polishing increases accuracy further

Evaluation of (polished) consensus haplotypes

How do variant counts change?

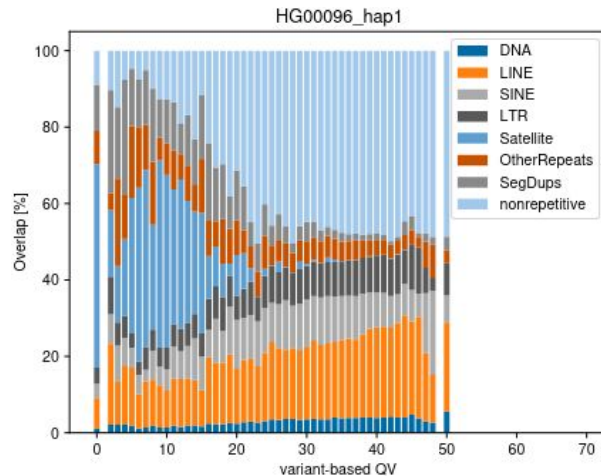
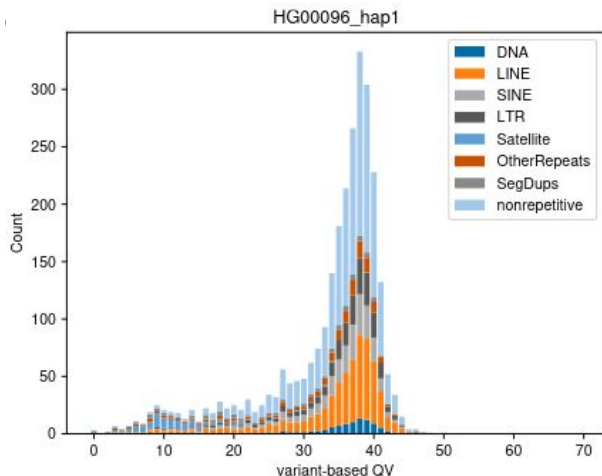
- For **10 evaluation samples**, jointly called variants across:
 - The two ground truth assemblies (HGSVC3)
 - The two unpolished haplotypes
 - The two polished haplotypes
- Check overlap with SVs in HPRC2 graph:
 - **Rare variants:** variants in assemblies **not** overlapping any graph SV ($\geq 50\text{bp}$)
 - **Ground truth variants:** all variants in assembly haplotypes
 - **Error variants:** variants in consensus HTs not present in assembly HTs



Less erroneous variant calls and more (rare) variants covered after polishing

Which regions are difficult to reconstruct?

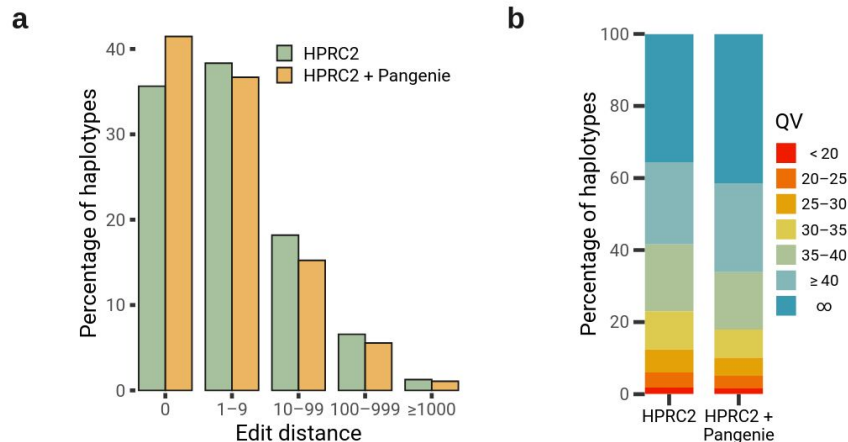
- Used variant-based QV estimation pipeline (HGSVC3 paper) to compute QVs for 1 Mbp windows
- Checked overlap of windows with BISER + RepeatMasker annotations



Low QV windows often overlap repetitive regions (especially Satellite/SINE)

Why are consensus haplotypes useful?

- Pangenome resource for much larger sample set
 - better capture human diversity across highly complex and polymorphic loci compared to “just” HPRC2 assemblies

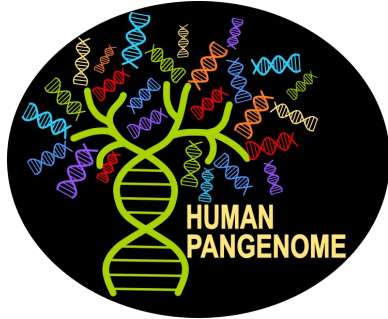


Timofey Prodanov

- Pangenome improves short-read based genotyping
 - Faster than alignment-based methods (~12 CPU hours per sample)
 - Able to access variants in regions not accessible by other short-reads methods
- Polishing improves QVs of consensus haplotypes (more rare variants and less FPs)
- Especially repetitive regions (Satellite/SINE) are still difficult to polish



Acknowledgements



Practical Session: PanGenie

EMBO Practical Course 2026, Naples



Jana Ebler

PostDoc @ Marschall Lab

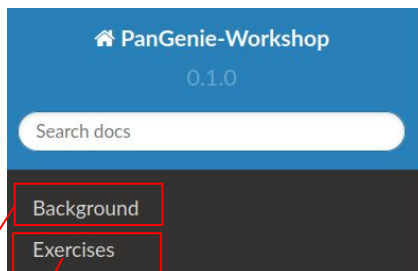
Heinrich Heine Universität Düsseldorf, Germany

PanGenie workshop

Data files can be found in the EMBO github repo:

<https://github.com/pangenome/embo-pangenomics-2026/tree/main/teaching/day6-jana-ebler>

The workshop is available here: <https://pangenie-workshop.readthedocs.io/>



🏠 / PanGenie workshop

[View page source](#)

PanGenie workshop

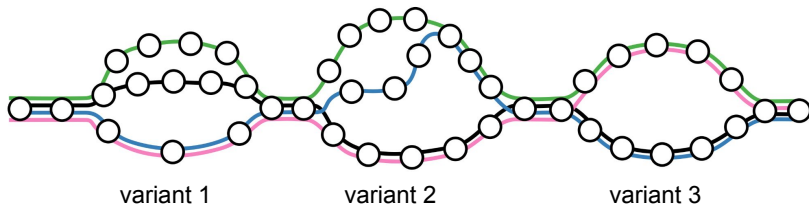
This workshop teaches the basics of PanGenie on a small use case. We provide background

Exercises we will be working on today

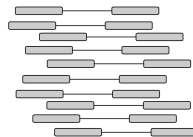
Details on PanGenie input data and commands

Input:

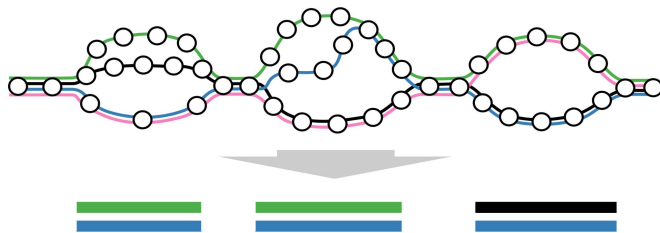
1. **Pangenome graph** constructed from known haplotypes



2. **Short reads** of the sample to be genotyped
(not in graph!)



Goal: Determine genotype of unknown genome for all variants contained in graph

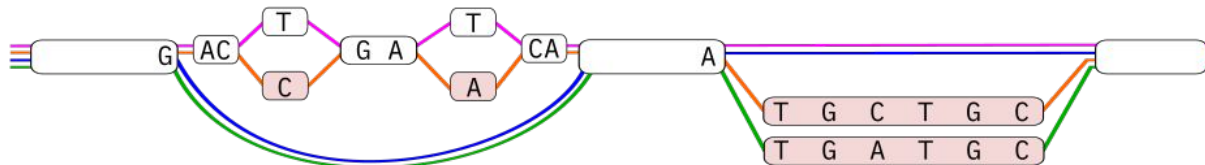


- **Reads**
 - WGS Illumina sequencing reads of a diploid individual
 - **FASTA** or **FASTQ** format (unaligned)
- **Reference genome**
 - reference genome in **FASTA** format
- **Pangenome reference**
 - pangenome reference represented in terms of a **VCF** file

Pangenome reference

- graphs build from haplotype-resolved assemblies
- variants produce **bubbles** in graph
- haplotypes represented as **paths**

pangenome graph



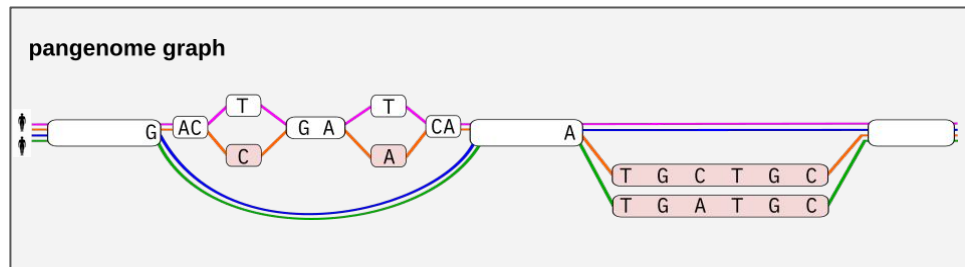
We use VCF format to encode the pangenome

```
##fileformat=VCFv4.2
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NA00001 NA00002 NA00003
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20 1234567 microsat1 GTC G,GTCT 50 PASS NS=3;DP=9;AA=G GT:GQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```

- Header lines starting with #
- Data lines
 - encode genetic variants relative to a reference genome
 - specify position, reference allele (REF), alternative alleles (ALT) and genotype information for samples
 - genotypes encode presence or absence of alleles on sample's haplotypes
 - “[” : alleles phased (we know which haplotype carries which allele), “/” : alleles unphased (we don't know which haplotype carries which allele)

VCF representation of graph

- Reference genome as additional path
- line in VCF = top-level bubble in graph
- REF field encodes reference sequence
- ALT encodes all non-reference paths
- sample columns encode haplotypes paths

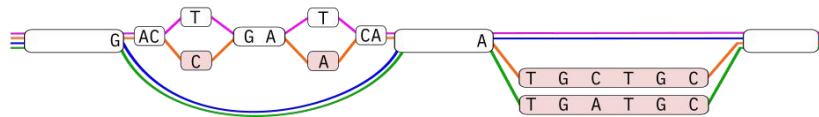


VCF records (multi-allelic representation)

CHROM	POS	REF	ALT	INFO	sample1	sample2
chr1	40	GACTGATCA	GACCGAACA, G	ID=chr1-43-SNP1:chr1-46-SNP2, chr1-40-DEL1	0 1	2 2
chr1	60	C	ATGCTGC, ATGATGC	ID=chr1-60-INS1, chr1-60-INS2	0 1	2 0

Representing nested variation

pangenome graph



Bubble VCF (PanGenie input)

- multi-allelic VCF
- INFO/ID annotations encoding nested variants
- each (nested) allele gets a unique ID

VCF records (multi-allelic representation)

CHROM	POS	REF	ALT	INFO	sample1	sample2
chr1	40	GACTGATCA	GACCGAACA, G	ID=chr1-43-SNP1:chr1-46-SNP2, chr1-40-DEL1	0 1	2 2
chr1	60	C	ATGCTGC, ATGATGC	ID=chr1-60-INS1, chr1-60-INS2	0 1	2 0

Callset VCF

- biallelic VCF
- each allele ID gets its own record
- makes comparison to other callsets possible

VCF records (bi-allelic representation)

CHROM	POS	REF	ALT	INFO	sample1	sample2
chr1	40	GACTGATCA	G	ID=chr1-40-DEL1	0 0	1 1
chr1	43	T	C	ID=chr1-43-SNP1	0 1	0 0
chr1	46	T	A	ID=chr1-46-SNP2	0 1	0 0
chr1	60	C	ATGCTGC	ID=chr1-60-INS1	0 1	0 0
chr1	60	C	ATGATGC	ID=chr1-60-INS2	0 0	1 0

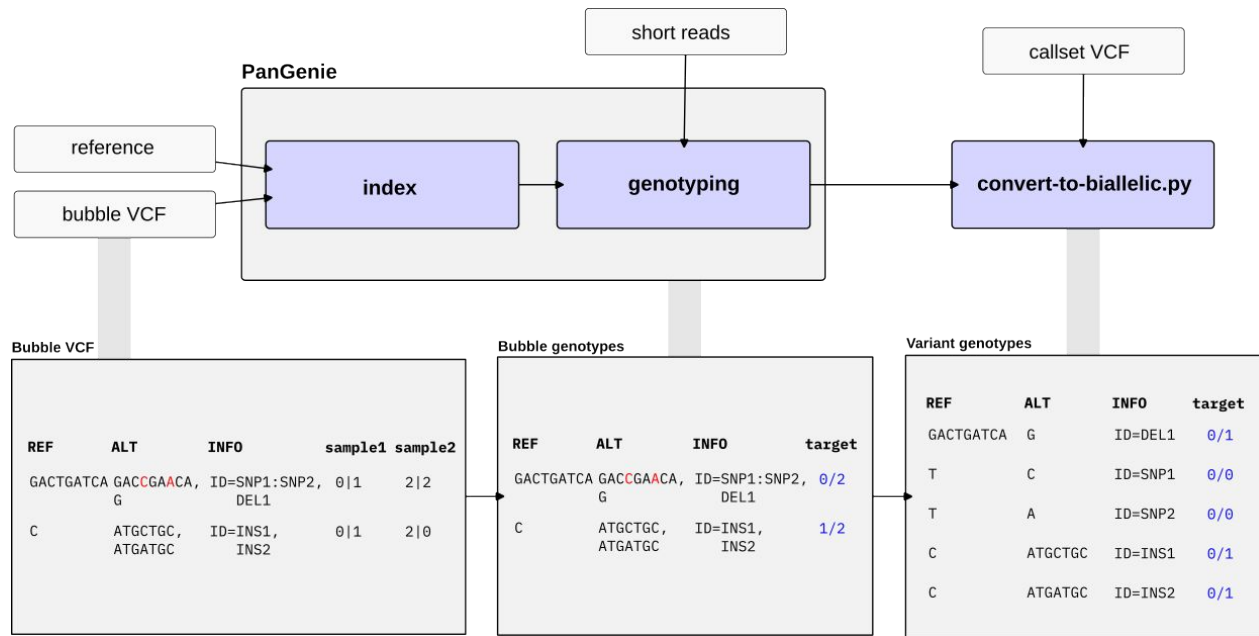
How to create such VCFs?

How to generate VCFs is not the focus of this workshop, but all information on this is available in the PanGenie Documentation (<https://pangenie.readthedocs.io>)

There we provide:

- pipelines to generate your own PanGenie input VCFs
- PanGenie-ready input VCFs for several HGSC and HPRC Minigraph-Cactus graphs

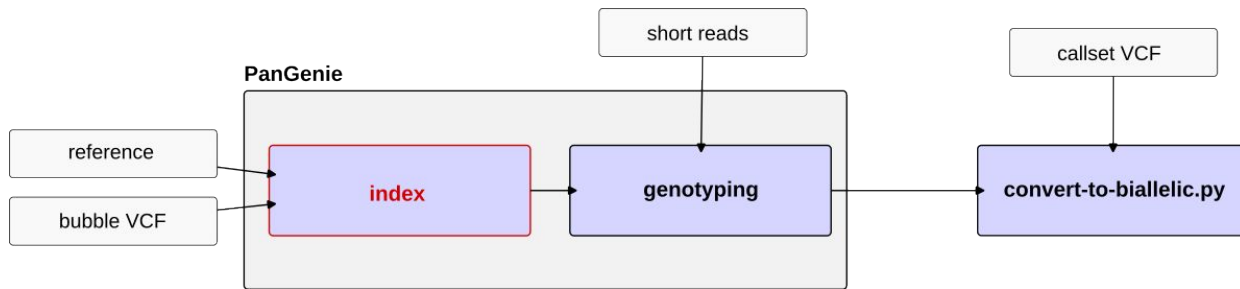
Focus today is to show how to use them with PanGenie



How to run PanGenie: indexing

- prepares data structures for genotyping
- determines unique k-mer sequences in the graph
- independent of sample data – needs to be **run only once**

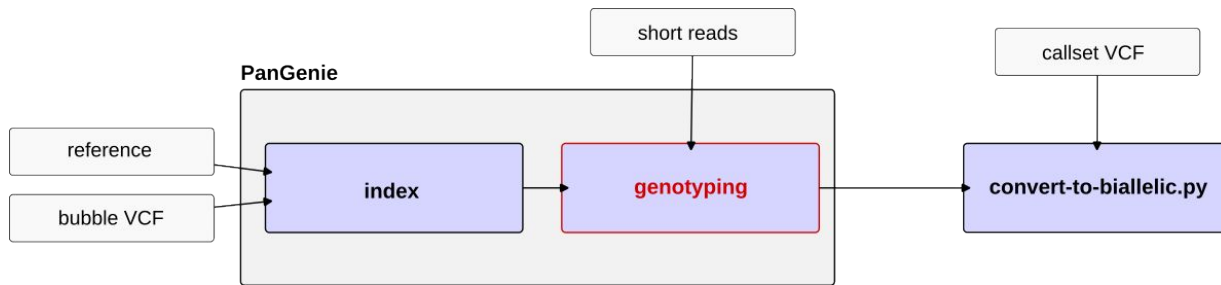
```
PanGenie-index -v <bubbles.vcf> -r <reference.fa> -t <number of threads> -o <index-prefix>
```



How to run PanGenie: genotyping

- count unique k-mers in the sample's sequencing data
- estimate average k-mer coverage
- genotype sample based on k-mer counts and haplotype paths in graph

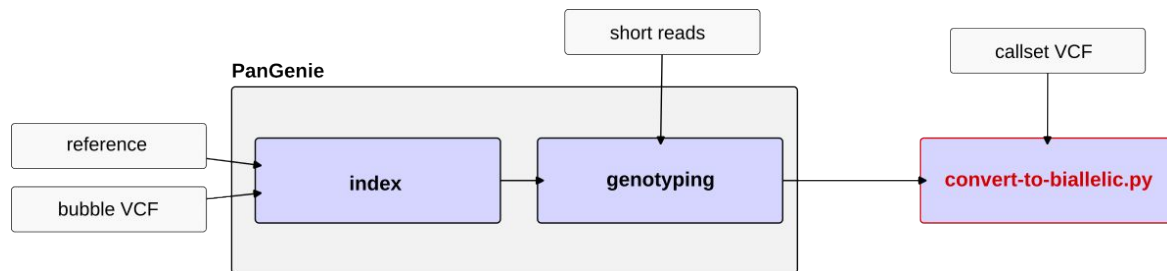
```
PanGenie -f <index-prefix> -i <reads.fa/fq> -s <sample-name> -j <nr threads kmer-counting> -t <nr threads genotyping> -o <genotyping-prefix>
```



Convert bubble genotypes to variant genotypes

- Additional postprocessing step of PanGenie bubble genotypes
- Only possible with VCFs containing special annotations in INFO/ID field
- Converts bubble genotypes to genotypes for all (nested) variant alleles

```
cat <genotyping-prefix>_genotyping.vcf | python3 convert-to-biallelic.py <callset.vcf> > <genotyping-prefix>_genotyping_biallelic.vcf
```



Exercises

Preparation & Inspecting the input data

- 1.1. Inspect the files and determine which file corresponds to the *bubble VCF* and which one to the *callset VCF*.
- 1.2. How many bubbles are in the underlying pangenome graph?
- 1.3. How many variant alleles are in the pangenome?
- 1.4. Of how many haplotypes was the graph constructed, i.e. how many paths are stored in the graph?

Hint: The commandline tool *bcftools* can be useful here.

<https://pangenie-workshop.readthedocs.io/>

Solutions: Preparation & Inspecting the input data

1.1. Inspect the files and determine which file corresponds to the *bubble VCF* and which one to the *callset VCF*?

Solutions: Preparation & Inspecting the input data

1.1. Inspect the files and determine which file corresponds to the *bubble VCF* and which one to the *callset VCF*?

Solution:

- Bubble VCF: `panel_multi_chr5:50200000-50400000.vcf.gz`
- Callset VCF: `panel_bi_chr5:50200000-50400000.vcf.gz`
- Reference genome: `reference_chr5:50200000-50400000.fasta`
- Reads:
 - NA19189: `NA19189_chr5:50200000-50400000.fasta`
 - NA19190: `NA19190_chr5:50200000-50400000.fasta`
 - NA19191: `NA19191_chr5:50200000-50400000.fasta`

Solutions: Preparation & Inspecting the input data

1.2. How many bubbles are in the underlying pangenome graph?

Solutions: Preparation & Inspecting the input data

1.2. How many bubbles are in the underlying pangenome graph?

Solution

We can determine the number of bubbles in the graph by counting the number of records in the bubble VCF, e.g. using this command:

```
bcftools view -H panel_multi_chr5:50200000-50400000.vcf.gz | wc -l
```

which will tell us that the number of bubbles is: **3,924**

Solutions: Preparation & Inspecting the input data

1.3. How many variant alleles are in the underlying pangenome graph?

Solutions: Preparation & Inspecting the input data

1.3. How many variant alleles are in the underlying pangenome graph?

Solution

The number of variant alleles can be determined by counting the number of records in the *callset VCF*, using the same command as for 1.1:

```
bcftools view -H panel_bi_chr5:50200000-50400000.vcf.gz | wc -l
```

This returns: **5,163**.

Solutions: Preparation & Inspecting the input data

1.4. Of how many haplotypes was the graph constructed, i.e. how many paths are stored in the graph?

Solutions: Preparation & Inspecting the input data

1.4. Of how many haplotypes was the graph constructed, i.e. how many paths are stored in the graph?

Solution

The number of haplotypes in the graph can be determined from the sample columns present in the *bubble* and *callset* VCFs (both contain the same samples). Since we are dealing with diploid samples, we can simply count the number of samples in the VCF and multiply the resulting number by two. One way to do this is by using the command:

```
bcftools query -l panel_multi_chr5:50200000-50400000.vcf.gz | wc -l | awk '{print $1 * 2}'
```

The number of haplotype paths is: **462**.

VCF records (multi-allelic representation)

CHROM	POS	REF	ALT	INFO	sample1	sample2
chr1	40	GACTGATCA	GACCGACA, G	ID=chr1-43-SNP1:chr1-46-SNP2, chr1-40-DEL1	0 1	2 2
chr1	60	C	ATGCTGC, ATGATGC	ID=chr1-60-INS1, chr1-60-INS2	0 1	2 0

Our dataset contains read data for three human individuals: NA19191, NA19189 and NA19190. We now want to genotype all three of them using PanGenie.

- 2.1. Create the PanGenie index structure for our dataset. Make sure which VCF file to use for this.
- 2.2. Genotype all three samples using the created index and the sample' sequencing data. Make sure to store the log output of the PanGenie command.
- 2.3. For all three samples, convert bubble genotypes to variant genotypes using the *convert-to-biallelic.py* script.
- 2.4. Inspect the log output of the commands run in 2.2. Which average k-mer coverages were computed by PanGenie for the three samples?
- 2.5. Combine the VCFs resulting from 2.3 into a single VCF using `bcftools merge`.

Note: for this workshop **only**, please add commandline parameter `-e 100000` to all PanGenie commands to save RAM.

<https://pangenie-workshop.readthedocs.io/>

Solutions: Running PanGenie

2.1. Create the PanGenie index structure for our dataset. Make sure which VCF file to use for this.

Solutions: Running PanGenie

2.1. Create the PanGenie index structure for our dataset. Make sure which VCF file to use for this.

Solution

We need to use the *bubble VCF* to generate an index with PanGenie. The following command needs to be run:

```
PanGenie-index -v panel_multi_chr5:50200000-50400000.vcf -r  
reference_chr5:50200000-50400000.fasta -o genotypes -t 1 -e 100000 &> indexing.log
```

Note: given that our dataset is very small, we can run PanGenie with a single thread (`-t 1`) and parameter `-e 100000`. For a real, whole-genome data set (PanGenie's use case), remove the `-e` parameter and increase the number of threads if possible.

Solutions: Running PanGenie

2.2. Genotype all three samples using the created index and the sample' sequencing data. Make sure to store the log output of the PanGenie command.

Solutions: Running PanGenie

2.2. Genotype all three samples using the created index and the sample' sequencing data. Make sure to store the log output of the PanGenie command.

Solution

Using the index previously computed, we can now genotype all three samples using their respective read data with the commands below:

```
PanGenie -f genotypes -i NA19189_chr5:50200000-50400000.fasta -o NA19189_pangenie_multi -t 1 -j 1 -s NA19189 -e 100000 &> pangenie_NA19189.log
```

```
PanGenie -f genotypes -i NA19190_chr5:50200000-50400000.fasta -o NA19190_pangenie_multi -t 1 -j 1 -s NA19190 -e 100000 &> pangenie_NA19190.log
```

```
PanGenie -f genotypes -i NA19191_chr5:50200000-50400000.fasta -o NA19191_pangenie_multi -t 1 -j 1 -s NA19191 -e 100000 &> pangenie_NA19191.log
```

Note: given that our dataset is very small, we can run PanGenie with a single thread (-t 1 and -j) and parameter -e 100000. For a real, whole-genome data set (PanGenie's use case), remove the -e parameter and increase the number of threads if possible.

Solutions: Running PanGenie

2.3. For all three samples, convert bubble genotypes to variant genotypes using the *convert-to-biallelic.py* script.

Solutions: Running PanGenie

2.3. For all three samples, convert bubble genotypes to variant genotypes using the *convert-to-biallelic.py* script.

Solution

Using the *callset VCF* and the genotype VCFs we generated in the previous task for all three samples, we can convert the bubble genotypes to variant genotypes with the commands shown below:

```
cat NA19189_pangenie_multi_genotyping.vcf | python3 convert-to-biallelic.py panel_bi_chr5:50200000-50400000.vcf.gz
| bgzip > NA19189_pangenie_bi_genotyping.vcf.gz
```

```
cat NA19190_pangenie_multi_genotyping.vcf | python3 convert-to-biallelic.py panel_bi_chr5:50200000-50400000.vcf.gz
| bgzip > NA19190_pangenie_bi_genotyping.vcf.gz
```

```
cat NA19191_pangenie_multi_genotyping.vcf | python3 convert-to-biallelic.py panel_bi_chr5:50200000-50400000.vcf.gz
| bgzip > NA19191_pangenie_bi_genotyping.vcf.gz
```

```
tabix -p vcf NA19189_pangenie_bi_genotyping.vcf.gz
```

```
tabix -p vcf NA19190_pangenie_bi_genotyping.vcf.gz
```

```
tabix -p vcf NA19191_pangenie_bi_genotyping.vcf.gz
```

Solutions: Running PanGenie

2.4. Inspect the log output of the commands run in 2.2. Which average k-mer coverages were computed by PanGenie for the three samples?

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Solution

We have stored the log outputs in files `pangenie_<sample>.log` (see 2.2.). These files contain a line starting with "Computed kmer abundance peak:" which provides the average k-mer coverage estimated for the respective sample.

We get the following numbers:

- NA19189: 25
- NA19190: 31
- NA19191: 26

Solutions: Running PanGenie

2.5. Combine the VCFs resulting from 2.3 into a single VCF using `bcftools merge`.

Solutions: Running PanGenie

2.5. Combine the VCFs resulting from 2.3 into a single VCF using `bcftools merge`.

Solution

We can use the following command to combine all three output VCFs into a single one:

```
bcftools merge NA19189_pangenie_bi_genotyping.vcf.gz
NA19190_pangenie_bi_genotyping.vcf.gz NA19191_pangenie_bi_genotyping.vcf.gz -Oz -o
merged-genotypes_bi.vcf.gz

tabix -p vcf merged-genotypes_bi.vcf.gz
```

The three individuals that we genotyped are related: NA19191 is the son of NA19189 (father) and NA19190 (mother). In this section we will analyze the genotypes further. One particularly interesting variant is chr5-50300172-INS->157412165>157412166>157412167-186. This variant is located in an ENCODE enhancer for B cell lymphomas and is associated with reduced expression of the immunoglobulin superfamily gene embigin (EMB), as reported by the HGSVC.

3.1. Check the genotypes of variant *chr5-50300172-INS->157412165>157412166>157412167-186* reported for all three individuals with PanGenie. From which parent was the variant inherited to the son?

3.2. Check the mendelian consistency for all variants (hint: you can use the command `bcftools +mendelian2`). How many variants are mendelian consistent? How many mendelian errors are there?

Solutions: Analyzing the genotypes

3.1. Check the genotypes of variant *chr5-50300172-INS->157412165>157412166>157412167-186* reported for all three individuals with PanGenie. From which parent was the variant inherited to the son?

Solutions: Analyzing the genotypes

3.1. Check the genotypes of variant *chr5-50300172-INS->157412165>157412166>157412167-186* reported for all three individuals with PanGenie. From which parent was the variant inherited to the son?

Solution

The variants in our VCFs are all assigned an ID in the INFO column of the VCF. All we need to do is to look for the ID in the VCF. The mother (NA19190) has genotype 1/1, the father (NA19189) has genotype 0/0 and the child is genotyped as 0/1. Therefore, the variant allele was inherited from the mother to the child.

Solutions: Analyzing the genotypes

3.2. Check the mendelian consistency for all variants (hint: you can use the command `bcftools +mendelian2`). How many variants are mendelian consistent? How many mendelian errors are there?

Solutions: Analyzing the genotypes

3.2. Check the mendelian consistency for all variants (hint: you can use the command `bcftools +mendelian2`). How many variants are mendelian consistent? How many mendelian errors are there?

Solution

We can run `bcftools mendelian` as:

```
bcftools +mendelian2 merged-genotypes_bi.vcf.gz -p 1X:NA19191,NA19189,NA19190 &>
mendelian-stats.txt
```

There are 5160 mendelian consistent variant sites and one mendelian error. The erroneous variant is chr5-50221495-SNV->157406351>157406353>157406354-1. Father 0/1, Mother 1/1, Child 0/0

Two of the variants are not in the genotyped set because PanGenie ignores variants that are less than two times the k-mer size away from the start or end of the genotyped region.